

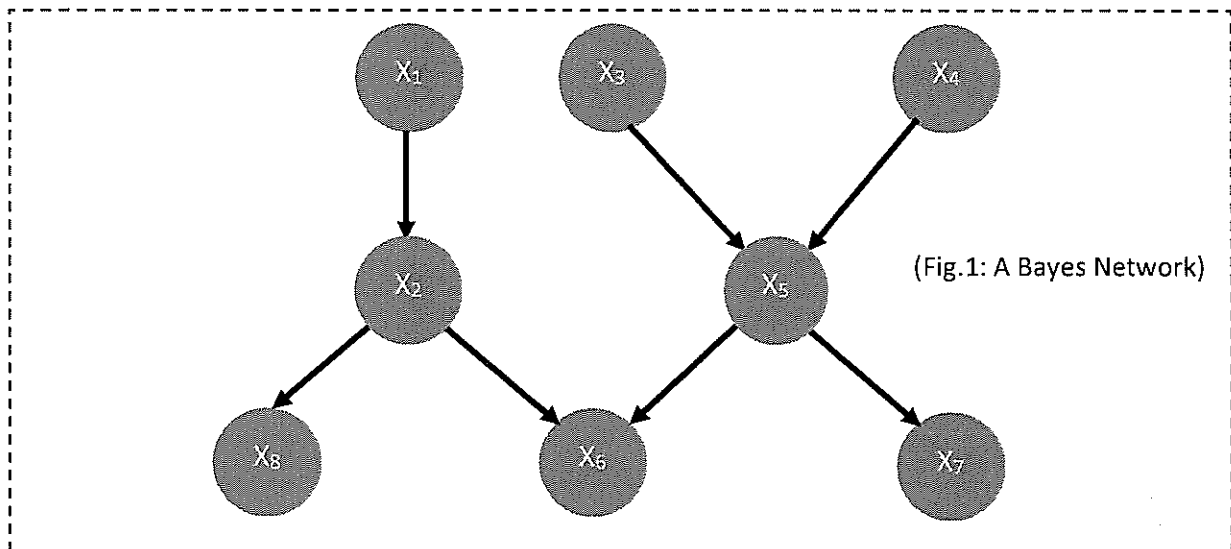
**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
HYDERABAD CAMPUS**

**Second Semester 2023-2024 Comprehensive Examinations
BIFS F464: Machine Learning**

Dt: 15/05/2024, Type: Closed Book, Weightage: 40%, Duration: 3 hrs

Note: There are 8 questions and 4 pages in the question paper. Write your answers **serially** and legibly. All the answers should be written in the answer sheet provided. Calculators (scientific and non-programmable) are allowed. Assume suitable data, if necessary. There is negative marking in Q.5.

Q.1 a) Answer the following considering the below Bayes network:



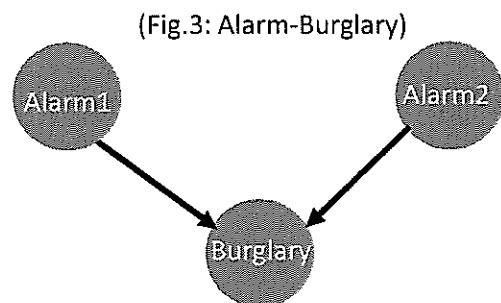
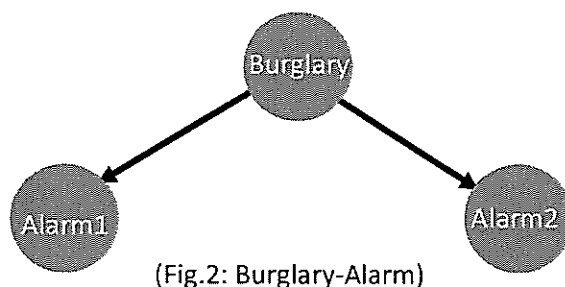
i) Given x_1 , what variable or variables are conditionally independent of x_7 ?

ii) Given x_5 , what variable or variables are conditionally independent of x_7 ?

iii) What is the conditional probability distribution for this Bayes network?

[2+2+2]

b) Assume that to safeguard your house, you recently installed two different alarm systems by two different manufacturers that use completely different sensors for their alarm systems. Alarm1, Alarm2, and Burglary nodes represent their happenings like alarm ringing and burglary taking place. Which one of the two Bayesian networks given below makes independent assumptions that are not true? Explain all the reasons possible.



[3 Marks]

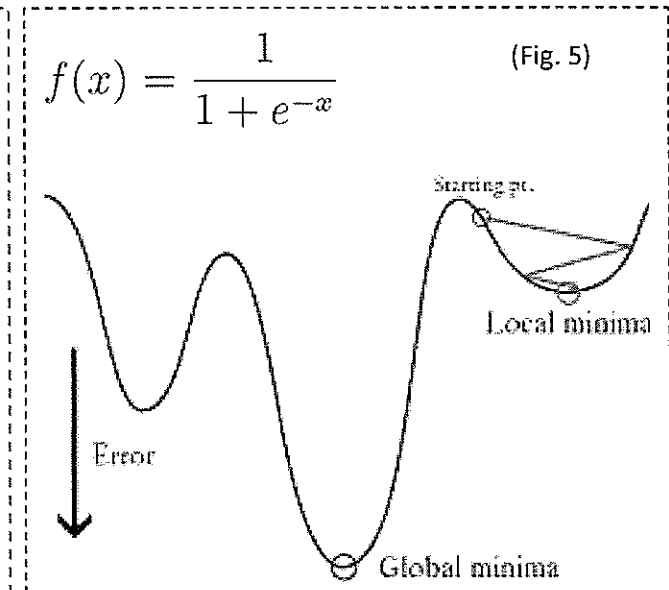
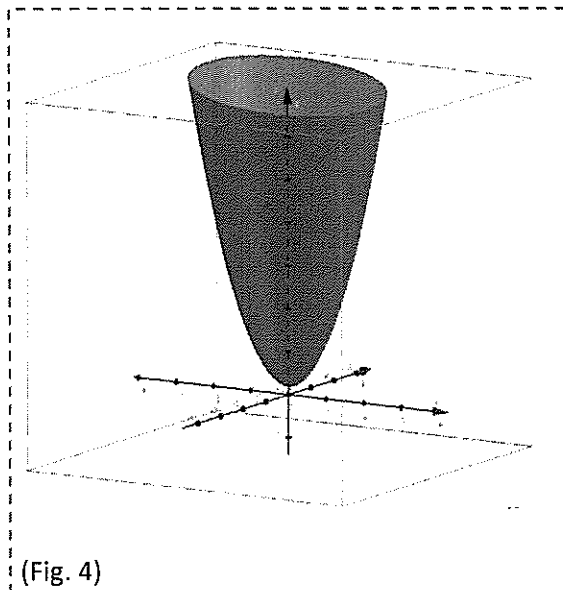
Q.2 a) For the below multi-variate function, find out its' gradient:

$$f(x, y) = x^2y + x$$

(P.T.O)

b) As discussed in the class training can be posed as an optimization problem, in which we optimize a cost function with respect to weights. Explain the principle of gradient descent algorithm using the below multi-variate function whose graph is also plotted in Fig. 4.

$$f(x, y) = x^2 + xy + y^2 + 0.2$$



c) Logistic regression discussed in the class makes use of a widely used logistic function for binary classification tasks as shown in Fig. 5. Using Mean Squared Error (MSE) as the loss function for logistic regression can lead to many problems. One such issue is shown in Fig.5 where a Gradient-based optimization algorithm, such as gradient descent, may struggle to converge to the global minimum, especially if the initial parameters are far from the optimal solution or if the optimization landscape is complex. What could be the reason for a non-convex loss function in this case? How could someone overcome from this non-convexity challenge? Explain briefly.

[3+3+3]

Q.3 a) Given the below training examples (Table 1) for a Decision Tree (ID3) learning algorithm, what is the Information Gain (I.G) of attribute A? Show your calculations neatly. Your values should be correct upto 4 decimal places.

No.	A	B	Class
1	T	T	Positive
2	T	T	Positive
3	T	F	Negative
4	F	F	Positive
5	F	T	Negative
6	F	T	Negative
7	T	F	Negative

(Table 1)

	Outlook	Temp	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes

(Table 2)

b) Table 2 describes a dataset to predict whether to play tennis or not on a particular day. Use candidate elimination algorithm studied in the course to show 'S' and 'G' sets after every example is presented to the learner.

[4+5]

Q.4 a) Given below (Table 3) is a dataset with built-up area in square feet and cost in INR of 3 houses. Perform Principal Component Analysis (PCA) to reduce these two features into One Principal Component. Show your calculations neatly.

Area	1500	2000	1200
Cost	300000	400000	200000

[9 Marks]

(P.T.O)

b) A Gaussian Mixture Model (GMM) is a probabilistic model used for representing the presence of subpopulations within an overall population. It assumes that the data is generated from a mixture of several Gaussian distributions, each associated with a different subpopulation. GMM works by iteratively estimating the parameters of these Gaussian distributions (mean and covariance) and the mixing coefficients (weights) that determine the contribution of each Gaussian component to the overall distribution. Write down the mathematical formulation of Gaussian Mixture Model.

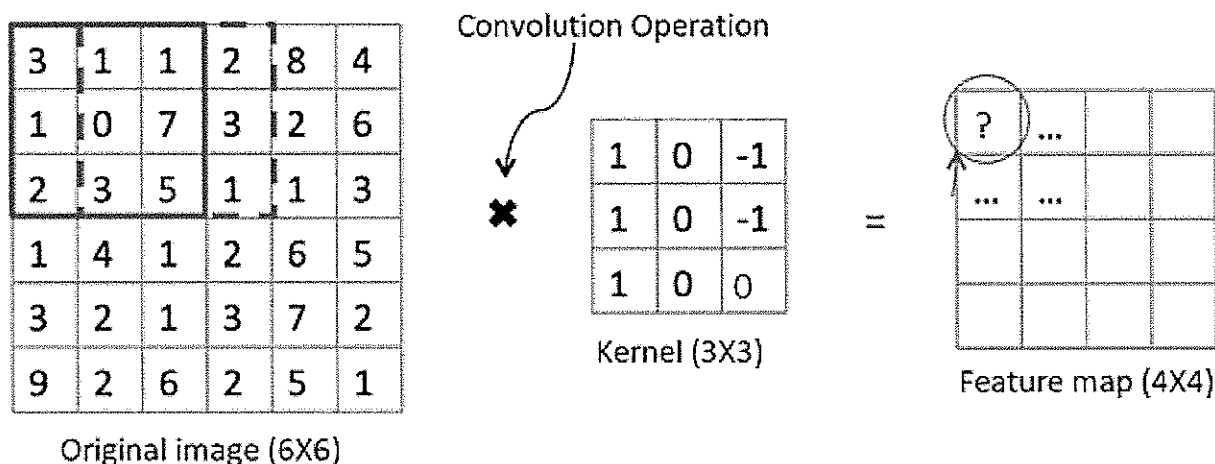
[4 Marks]

Q.5 a) Are these statements true or false? Give the reason for your answer. Each correct answer carries 1 Mark. For a wrong answer, -1 Mark will be awarded.

- The accuracy of a model is the training set accuracy, and its estimator is the test set accuracy.
- A larger training set would provide a lower variance estimate of the accuracy of a learned model.
- In cross validation, we evaluate the performance of an individual learned hypothesis.
- We usually train a discriminative model by maximizing the posteriors for true labels for supervised tasks.
- Naïve Bayes assumes conditional independence of features to decompose the joint probability into the conditional probabilities.
- The principal component with the largest eigenvalue maximizes the reconstruction error.

[6 Marks]

b) What is share structure property in a Convolutional Neural Network (CNN)? Given below (Fig. 6) is an example of Convolution operation. Find out the value marked as? in the figure. Show your calculations.



(Fig. 6)

[2 + 1 Marks]

c) In a CNN spatial variations in the input (such as translation, or scaling) are tolerated to some extent, making the network more robust to variations in input data. Which layer in the CNN architecture is responsible for Spatial Invariance and why? Explain.

[1 Mark]

Q.6 a) For the Recurrent Neural Network (RNN) shown in Fig. 7 (next page) where the hidden states are updated recurrently based on the current input and the previous hidden state, use Backpropagation Through Time training algorithm to find out the following. Use Sigmoid as the activation function. Assume the loss function to be the Mean Squared Error as given below:

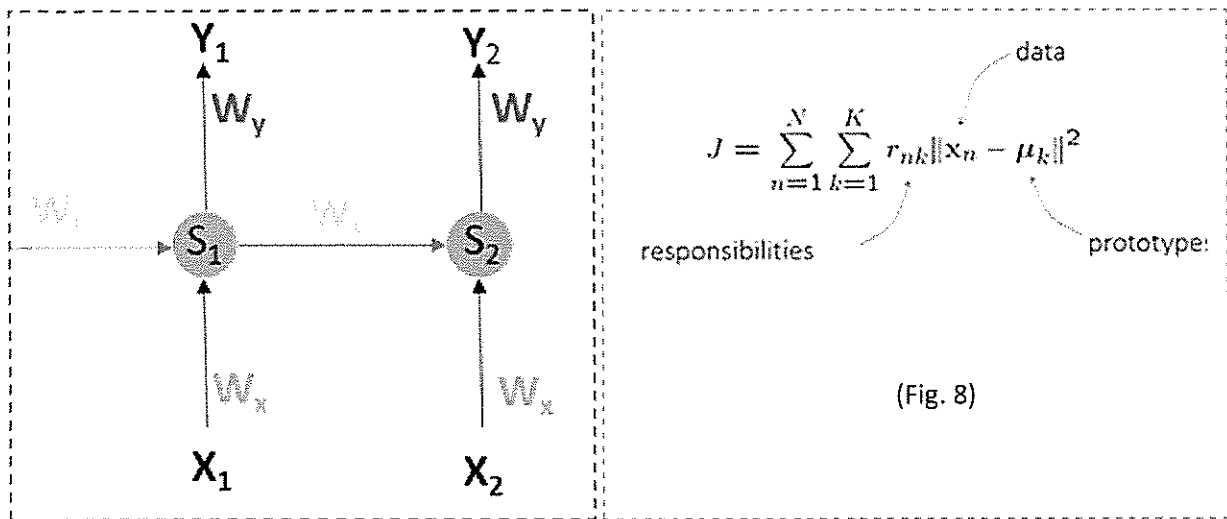
$$E = (1/N) \sum_{i=1}^N (Y_i - \hat{Y}_i)^2 \quad (\partial E_2 / \partial W_y) = ? \quad (\partial E_2 / \partial W_s) = ? \quad \text{and} \quad (\partial E_2 / \partial W_x) = ?$$

[4 Marks]

b) RNNs are a class of neural networks designed to work with sequential data by maintaining internal memory states. LSTMs are a specific type of RNN architecture that address some of the limitations of traditional RNNs, such as the vanishing gradient problem. How does LSTM do this? Explain.

[3 Marks]

(P.T.O)



(Fig. 7: A Recurrent Neural Network)

c) The research paper titled "Traffic Graph Convolutional Recurrent Neural Network ..." by Zhiyong Cui discussed in the class was a ConvLSTM in which authors identified Network wide traffic states with most influential roadways. Write in your own words why is it called ConvLSTM and not Conv+LSTM? How did they use the Convolution operator in their paper? Briefly explain.

[3 Marks]

Q.7 a) A binary classifier with labels $y \in \{1 \text{ (positive)}, 0 \text{ (negative)}\}$ was evaluated using 2000 examples in which 40% of all examples are negative. It was found that the classifier has 70% sensitivity and 62% accuracy. What is its' Confusion Matrix? Show your calculations. (Note: Sensitivity is a measure of how well a machine learning model can detect positive instances).

[4 Marks]

b) Given four points in a 2-D plane, $\{(2,3), (3,5), (6,2), (8,3)\}$. Show the run of K-Means Clustering algorithm manually that you studied in the course to cluster them to two clusters. Show the clusters finally after the convergence. The distortion measure that we discussed is as given in Fig. 8. How do you determine μ_k keeping r_{nk} fixed? Describe.

[3+1 Marks]

c) Suppose you have a dataset containing two classes, labelled as positive (+1) and negative (-1). You decide to use a Support Vector Machine (SVM) classifier with a linear kernel to classify this data. How does the SVM algorithm find the optimal decision boundary? Explain.

[2 Marks]

Q.8 a) Consider an autonomous vehicle navigating in a grid world. The vehicle can move up, down, left, or right in each time step, but it may encounter obstacles that block certain grid cells. The vehicle's goal is to reach a specified destination grid cell while avoiding obstacles. Define what a Markov Decision Process (MDP) is and explain its components in the context of this problem.

[4 Marks]

b) Explain the concepts of bagging and boosting in the context of ensemble learning. How are these techniques utilized in the construction of a random forest algorithm? Compare and contrast bagging and boosting, highlighting their respective advantages and disadvantages. Additionally, discuss why bagging is preferred over boosting in the context of random forests.

[3 Marks]

c) Decision trees are typically evaluated using metrics like accuracy, Gini impurity, or entropy. Log loss is more commonly used in evaluating models like logistic regression or neural networks. However, if someone decides to use log loss metric to evaluate decision trees how is it feasible? (Hint: You may think of a Decision Tree or use of Decision Trees in building a better model)

[3 Marks]

----- End of the text! -----